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Lab Week 4

1. Ocean.jpg

ExifTool Output	
Structured ExifTool output grouped for easier reading on phones and smaller screens.	
Text 50 tags	
ExifTool 1 tags	
ExifToolVersion	13.42
System 7 tags	
FileName	ocean.jpg
Directory	/
FileSize	42860
FileModifyDate	2025:10:27 16:41:41+00:00
FileAccessDate	2025:10:27 16:41:41+00:00
FileInodeChangeDate	2025:10:27 16:41:41+00:00
FilePermissions	100000
File 10 tags	
FileType	JPEG
FileTypeExtension	JPG
MIMEType	image/jpeg
Comment	THIS IS THE HIDDEN FLAG
ImageWidth	640

An analysis of ocean.jpg was conducted to detect any flags hidden within its background data. Using the online metadata viewer exif.tools, the file's metadata was inspected. The examination revealed a specific field labeled 'Comment,' which explicitly contained the string 'THIS IS THE HIDDEN FLAG,' marking the initial discovery.

2. dog.jpg

```
kali@kali: ~/Desktop/image/_dog.jpg.extracted
Session Actions Edit View Help

(kali@kali)~$ cd image
(kali@kali)~/Desktop/image$ binwalk dog.jpg

DECIMAL    HEXADECIMAL  DESCRIPTION
-----
0          0x0          JPEG image data, JFIF standard 1.01
88221     0x1589D      Zip archive data, at least v1.0 to extract, com
pressed size: 21, uncompressed size: 21, name: hidden_text.txt
88384     0x15940      End of Zip archive, footer length: 22

(kali@kali)~/Desktop/image$ binwalk -e dog.jpg

DECIMAL    HEXADECIMAL  DESCRIPTION
-----
88221     0x1589D      Zip archive data, at least v1.0 to extract, com
pressed size: 21, uncompressed size: 21, name: hidden_text.txt

WARNING: One or more files failed to extract: either no utility w
it's unimplemented

(kali@kali)~/Desktop/image$ ls
computer.jpg dog.jpg-0.extracted dog.jpg.extracted rubiks.jpg
dog.jpg _dog.jpg-1.extracted ocean.jpg    solitaire.exe

(kali@kali)~/Desktop/image$ cd _dog.jpg.extracted
(kali@kali)~/Desktop/image/_dog.jpg.extracted$ ls
1589D.zip hidden_text.txt

(kali@kali)~/Desktop/image/_dog.jpg.extracted$ cat hidden_text.txt
THIS IS A HIDDEN FLAG

(kali@kali)~/Desktop/image/_dog.jpg.extracted$
```

In the next task, dog.jpg was analyzed using the command line on a Kali Linux machine. An initial scan using the binwalk command revealed an embedded Zip archive and a file named hidden_text.txt. To access the contents, binwalk -e was executed to extract the hidden files into a new directory named _dog.jpg.extracted. After navigating into the folder and confirming the files using ls, the cat hidden_text.txt command was used to display the contents, successfully revealing the hidden flag: "THIS IS A HIDDEN FLAG"

3. Solitaire

```
(kali㉿ kali)-[~/Desktop/image/_dog.jpg.extracted]
$ cat hidden_text.txt
THIS IS A HIDDEN FLAG

(kali㉿ kali)-[~/Desktop/image/_dog.jpg.extracted]
$ file solitaire.exe
solitaire.exe: cannot open `solitaire.exe' (No such file or directory)

(kali㉿ kali)-[~/Desktop/image/_dog.jpg.extracted]
$ cd ..

(kali㉿ kali)-[~/Desktop/image]
$ file solitaire.exe
solitaire.exe: PNG image data, 640 x 449, 8-bit/color RGBA, non-interlaced

(kali㉿ kali)-[~/Desktop/image]
$ mv solitaire.exe solitaire.png

(kali㉿ kali)-[~/Desktop/image]
$ ristretto solitaire.png
```

For the third file, solitaire.exe, the objective was to verify whether it was a genuine executable. I executed the file command in the terminal to inspect its true signature. The output indicated that the file was actually PNG image data with a resolution of 640 x 449, proving that the .exe extension was merely a decoy. To correct this, I used the mv command to rename the file to solitaire.png. Finally, I opened it using the ristretto image viewer, which displayed the actual image of playing cards.



4. Rubiks

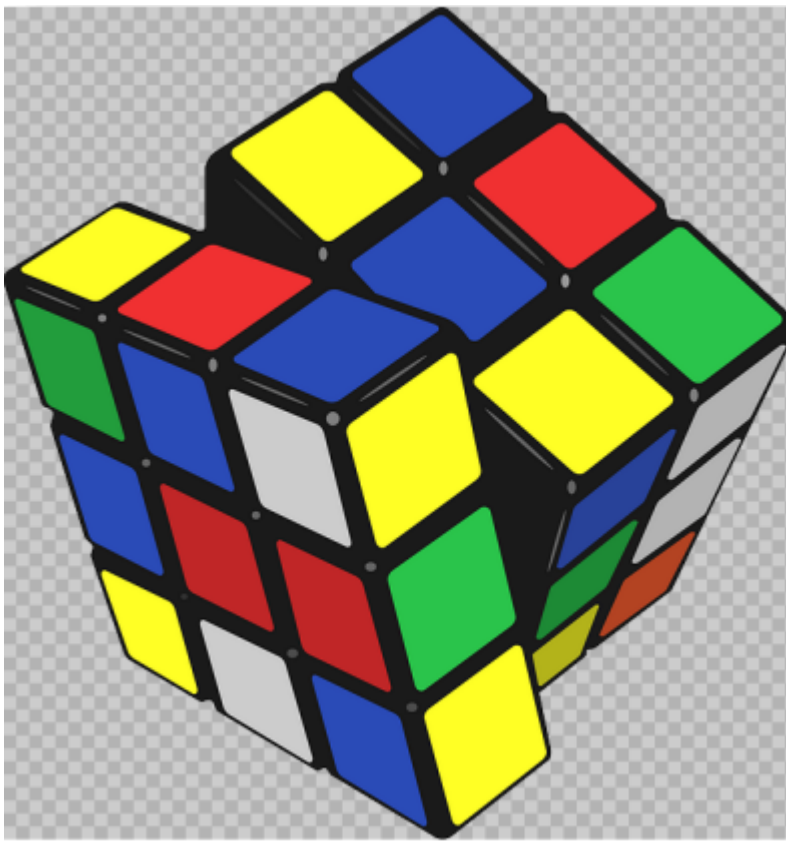
```
(kali㉿ kali) ~/Desktop/image
$ file rubiks.jpg
rubiks.jpg: PNG image data, 609 x 640, 8-bit/color RGBA, non-interlaced

(kali㉿ kali) ~/Desktop/image
$ mv rubiks.jpg rubiks.png
mv: cannot stat 'rubiks.jpg': No such file or directory

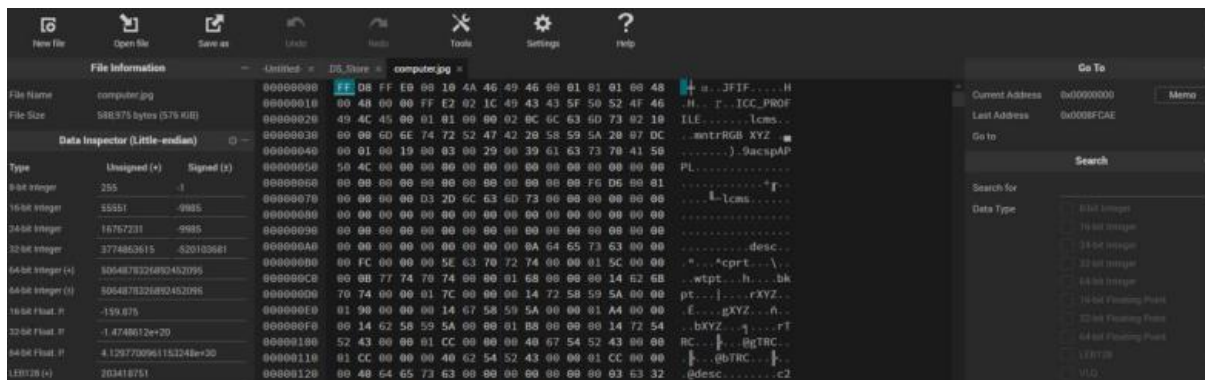
(kali㉿ kali) ~/Desktop/image
$ mv rubiks.jpg rubiks.png

(kali㉿ kali) ~/Desktop/image
$ ristretto rubiks.png
```

The analysis of rubiks.jpg followed a similar process to identify another extension mismatch. Running the file command confirmed that it was actually PNG image data rather than a JPEG. I executed the mv command to change the extension to .png to align with its true format. Once corrected, I opened the file with the ristretto image viewer, which displayed a 3D graphic of a Rubik's Cube.



5. computer.png



The final part of the lab involved inspecting the raw binary data of computer.jpg. To achieve this, I uploaded the file to hexed.it, a web-based hex editor, to analyze its file headers and hexadecimal values.



"To improve data readability, I also utilized the dCode Strings Extractor tool. Processing the file through this utility allowed me to extract human-readable text strings from the binary code, such as ICC_PROFILE and mnrRGB XYZ. This demonstrated how combining hex editing and string extraction can effectively analyze a file's internal structure.